MTH1004

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2022

Probability, Statistics, and Data

Module Leader: Dr Stefan Siegert

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Statistical Tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count specified

SECTION A

1. (a) The random variable X has an Exponential distribution, that is, its probability density function (pdf) is given by $f_X(x) = \lambda e^{-\lambda x}$ for $x > 0$ and $f_X(x) = 0$ for $x \leq 0$, where $\lambda > 0$ is the rate parameter.

   (i) Derive the cumulative distribution function of X from its pdf.

   (ii) Write down (or derive) the expectation of X.

   (iii) Waiting times at a taxi stop are modelled as independent, Exponentially distributed random variables. The expected waiting time is 5 minutes. What is the probability of having to wait more than 10 minutes for a taxi? (10)

   (b) (i) Write down the numerical value of $\sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x}$ where $p \in (0, 1)$ and $n > 0$.

   (ii) State the Binomial Theorem and explain how it can be used to justify your answer to question 1(b)i. (10)

   (c) Let $X \sim \text{Unif}(0, 2)$, where the pdf of X is $f_X(x) = 1/2$ for $0 < x < 2$, and let $Y = X^{-1/2}$.

   (i) Write down (or derive) the cumulative distribution function of X.

   (ii) Write down the range space of Y.

   (iii) Calculate the cumulative distribution function of Y.

   (iv) Calculate the probability density function of Y.

   (v) Calculate the expectation of Y. (12)

   (d) At a dinner party, every guest has a glass and clinks glasses once with each of the other guests. You hear 861 "ping" sounds. Calculate how many guests there are in the room. (6)

   (e) In chemistry, pH is a measure of the acidity of a liquid. The pH of water in a lake is known from past experience to vary naturally according to a Normal distribution with expectation $\mu = 6.5$ and standard deviation $\sigma = 0.1$. Following a pollution incident, a sample of 10 measurements was found to have mean $\bar{y} = 6.43$.

   (i) Conduct a test of size 1% of the null hypothesis that the expected pH of the water is 6.5 against the alternative hypothesis that the expected pH is not 6.5. State clearly your test statistic, its null distribution, the critical region and your conclusion about the acidity of the water.

   (ii) Calculate the p-value for this test. (12)

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SECTION B

2. Let $Y_1, \ldots, Y_n$ be independent random variables with $\mathrm{E}(Y_1) = 1/\lambda$ and $\mathrm{Var}(Y_1) = 1/\lambda^2$. For a positive constant λ_0, consider testing the null hypothesis $H_0 : \lambda = \lambda_0$ against the alternative hypothesis $H_1 : \lambda > \lambda_0$ using the test statistic

$$T = \sqrt{n}(\lambda_0 \bar{Y} - 1),$$

where $\bar{Y} = \sum_{i=1}^{n} Y_i/n$.

(a) Given a critical value, c, explain why it would be appropriate to reject H_0 when $T < c$ (rather than when $T > c$). (3)

(b) Explain why the null distribution of T is approximately $\mathrm{N}(0, 1)$ when n is large. (8)

(c) Show that, when $\lambda = \lambda_1$, the power of the test is approximately equal to

$$P\left[Z < \frac{c\lambda_1}{\lambda_0} + \sqrt{n}\left(\frac{\lambda_1}{\lambda_0} - 1\right)\right],$$

where $Z \sim \mathrm{N}(0, 1)$. (6)

(d) (i) Find the critical value, c, for which the size of this test is approximately 5%.

(ii) Using the value of c from part (i), find the sample size, n, for which the power of this test is at least 0.8 when $\lambda_1 = 1.1\lambda_0$. (8)

[25]

3. Let $Y_1, \ldots, Y_n$ be independent random variables, each with cumulative distribution function

$$P(Y_1 \leq y) = 1 - \left(1 - \frac{y}{\theta}\right)^{1/2} \qquad \text{for } 0 \leq y \leq \theta,$$

where θ is a parameter.

(a) You may use the following results to answer this part of the question: $\mathrm{E}(Y_1) = 2\theta/3$ and $\mathrm{Var}(Y_1) = \theta^2/45$.

(i) Write down the method of moments estimator for θ.

(ii) Calculate the bias and variance of this estimator. (6)

(b) Consider the estimator $\hat{\theta} = \max\{Y_1, \ldots, Y_n\}$.

(i) Find the cumulative distribution function of $\hat{\theta}$.

(ii) Calculate the bias of $\hat{\theta}$.

(iii) Calculate the limit of $P(|\hat{\theta} - \theta| > y)$ as $n \to \infty$, where $0 < y < \theta$. (19)

[25]

4. The random variable $X \in \{0, 1\}$ is the outcome of the flip of a fair coin with sides labelled 0 and 1. Conditional on a value of $X = x$, the random variable $Y \in (0, \infty)$ has a Gamma distribution with shape parameter $1+x$ and rate parameter 1. (The probability density function of a Gamma-distributed random variable Y with shape parameter α and rate parameter β is $f_Y(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$.)

(a) Write down (or derive) $E[Y|X = 0]$ and $E[Y|X = 1]$. (3)

(b) Calculate $E[Y]$. (8)

(c) Calculate the marginal probability density function of Y. (6)

(d) Calculate $P((X = 1) \cap (Y > 1))$. (8)

[25]